```
        return $n * factorial ($n - 1);
    }
}
echo factorial (4), "\n";
echo factorial (10), "\n";

?>
```

In this code example, we calculate the factorial of two numbers.

return $n * factorial ($n - 1);

Inside the body of the factorial function, we call the factorial function with a modified argument. The function calls itself.

```
$ php recursion.php
24
3628800
```

These are the results.

So what does the syntax for defining a function look like? Here is a simple example:




```
<?php
function addNumbers($num1, $num2) {
    $result = $num1 + $num2;
    return $result;
}
```

3.1.10 PHP global and local variables

Next we will talk about the scope of the variables in PHP. A *scope* is the range in which a variable can be referenced. When we work with functions, there are two basic scopes: the global and the local scope. The local scope is also called a function scope.